

 matsuo-lab/**weblab-10b-instruction-sft** 

 like 73

 Text Generation  Transformers  PyTorch gpt_neox

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 Text Generation

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Model tree for matsuo-lab/weblab-10b-instruction-sft

Quantizations

1 model

 Spaces using matsuo-lab/weblab-10b-instruction-sft 3

weblab-10b-instruction-sft

Overview

This repository provides a Japanese-centric multilingual GPT-NeoX model of 10 billion parameters.

- Library

The model was trained using code based on [EleutherAI/gpt-neox](#).

- Model architecture

A 36-layer, 4864-hidden-size transformer-based language model.

- **Pre-training**

The model was trained on around **600B** tokens from a mixture of the following corpora.

- [Japanese C4](#)
- [The Pile](#)

- **Instruction-supervised-finetuning**

The model was finetuned on a subset records from a mixture of the following dataset. Training epoch: 1.

- [Alpaca \(English\)](#).
- [Alpaca \(Japanese translation\)](#).
- [Flan 2021 \(English\)](#).
- [Flan CoT \(English\)](#).
- [Flan Dialog \(English\)](#).

- **Model Series**

Variant	Link
weblab-10b-instruction-sft	https://huggingface.co/matsuo-lab/weblab-10b-instruction-sft
weblab-10b	https://huggingface.co/matsuo-lab/weblab-10b

- **Authors**

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[Benchmarking](#)

- **Japanese benchmark : JGLUE 8-task (2023-08-27)**

- We used [Stability-AI/lm-evaluation-harness](#) library for evaluation.
- The 8-task average accuracy is based on results of JCommonsenseQA-1.1, JNLI-1.1, MARC-ja-1.1, JSQuAD-1.1, jaqket_v2-0.2, xlsun_ja-1.0, xwinograd_ja, and mgsm-1.0.
- model loading is performed with float16, and evaluation is performed with template version 0.3 using the few-shot in-context learning.
- The number of few-shots is 3,3,3,2,1,1,0,5.
- special_tokens_map.json is modified to avoid errors during the evaluation of the second half benchmarks. As a result, the results of the first half benchmarks became slightly different.

model	average	jcommonsenseqa	jnli	marc_ja	jsquad	jaqket_v2	xlsun_ja	xwinog
weblab-10b-instruction-sft	59.11	74.62	66.56	95.49	78.34	63.32	20.57	71.95
weblab-10b	50.74	66.58	53.74	82.07	62.94	56.19	10.03	71.95

• Japanese benchmark : JGLUE 4-task (2023-08-18)

- We used [Stability-AI/lm-evaluation-harness](#) library for evaluation.
- The 4-task average accuracy is based on results of JCommonsenseQA-1.1, JNLI-1.1, MARC-ja-1.1, and JSQuAD-1.1.
- model loading is performed with float16, and evaluation is performed with template version 0.3 using the few-shot in-context learning.
- The number of few-shots is 3,3,3,2.

Model	Average	JCommonsenseQA	JNLI	MARC-ja	JSQuAD
weblab-10b-instruction-sft	78.78	74.35	65.65	96.06	79.04

Model	Average	JCommonsenseQA	JNLI	MARC-ja	JSQuAD
weblab-10b	66.38	65.86	54.19	84.49	60.98

🔗 How to use the model

```
import torch
from transformers import AutoTokenizer, AutoModelForCausalLM

tokenizer = AutoTokenizer.from_pretrained("matsuo-lab/weblab-10b-instruction-sft")
model = AutoModelForCausalLM.from_pretrained("matsuo-lab/weblab-10b-instruction-sf

if torch.cuda.is_available():
    model = model.to("cuda")

text = "大規模言語モデルについて説明してください。"
text = f'以下は、タスクを説明する指示です。要求を適切に満たす応答を書きなさい。\\n\\n### 指示:\\n
token_ids = tokenizer.encode(text, add_special_tokens=False, return_tensors="pt")

with torch.no_grad():
    output_ids = model.generate(
        token_ids.to(model.device),
        max_new_tokens=100,
        do_sample=True,
        temperature=0.7,
        top_p=0.95
    )

output = tokenizer.decode(output_ids.tolist()[0])
print(output)
```

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